OSAMA KHALIL

SENIOR DEVOPS

DevOps Engineer | Microsoft Azure | Terraform | CI/CD | Linux Administration

Current: Senior Engineer - DevOps at Ericsson India Global Services | 11 Years IT (6 Years DevOps & Linux)

Location: New Delhi, India

Expertise: Azure Cloud, IaC & CI/CD

Education: B.Tech ECE + PG Cloud Computing

PROFESSIONAL CAREER TIMELINE (11 YEARS)

Nutek India Pvt. Ltd.
Site Engineer (Field) | Oct 2013 - Jan 2016
Telecom site infrastructure & Nokia BTS

Universal Telecom Pvt. Ltd.
BTS Engineer (NOC/BSS) | Mar 2016 - Jan 2017
NOC operations, BSS alarms & incident mgmt

Engicell Technology Ltd.
Telecom Implementation | Feb 2017 - Oct 2020
Airtel/Huawei: Zambia, Kenya, Malawi, Namibia

Emerald Telecom Pvt. Ltd.
Network Engineer (Linux) | Mar 2021 - Oct 2021
Linux gateways, iptables, routing, tcpdump

Ericsson India Global Services
Senior Engineer - DevOps | Dec 2021 - Present
Azure IaC, Terraform, CI/CD, Enterprise Linux

PRIMARY PROJECT: TELECOM NETWORK MANAGEMENT SYSTEM (NMS) ON AZURE

1. FRONTEND PRESENTATION TIER

Azure Storage Static Website ($web container)
Decoupled Single Page Application (SPA) for Telecom NMS

Client Ingress & Network Delivery
- Secure HTTPS / TLS 1.3 encryption (TCP Port 443)
- Real-time NMS dashboard delivery for Telecom NOC users

DevOps & Automation Highlights
- Provisioned via Terraform azurerm_storage_account
- Automated artifact sync in Azure DevOps Release Pipeline

2. BACKEND APPLICATION TIER (Azure VNet: snet-backend-app)

Linux Virtual Machine (Enterprise RHEL / Ubuntu)
- OS hardening, security patching, performance tuning & diagnostics
- Private IP: 10.0.2.10 | Secured with SSH keys & RBAC

Nginx Web Server & Reverse Proxy
- SSL/TLS Termination, rate limiting, gzip compression & access logs
- High-throughput reverse proxy pass to FastAPI daemon (localhost:8000)

FastAPI Telecom Application Daemon
- Asynchronous Python REST APIs for telecom network monitoring
- Managed via systemd service with automated health probe checks

3. DATABASE TIER (Azure VNet: snet-database)

Azure Database for PostgreSQL (Flexible Server)
Relational store for network topology, inventory & active alarm logs

Private Connectivity & Strict NSG Rules
- Zero Public IP | Private Endpoint / VNet Integration
- NSG Rule: Inbound Port 5432 strictly allowed from 10.0.2.0/24 ONLY

Enterprise Reliability & Governance
- Automated backups, zone redundancy & point-in-time recovery
- Credentials in Azure Key Vault; monitored via Prometheus & Monitor

TERRAFORM INFRASTRUCTURE AS CODE (IaC) LIFECYCLE

Modular Architecture: Reusable modules (vnet, subnets, nsg, linux_vm, postgresql, storage, key_vault)

Remote State Management: Hosted on Azure Blob Storage with automated lease locking across Dev/Stage/Prod

Shift-Left Security Scanning: Pre-commit & PR gates running Checkov, tfsec and tflint linters

Predictable Execution: terraform plan speculative delta review prior to automated deployment

Zero Configuration Drift: Declarative code repository acts as single source of truth for all Azure cloud infra

ENTERPRISE CI/CD PIPELINES (AZURE DEVOPS & GITHUB ACTIONS)

Branch Governance: Trunk-based strategy, feature branches, mandatory peer code reviews & branch protection

Automated Quality Gates: PR triggers fire automated pytest suites, flake8 linting and YAML validations

Multi-Stage Delivery: Automated Dev deployment -> Pre-production Stage validation -> Prod Manual Sign-off

Application Packaging: Immutable drop artifacts published to Azure Pipeline Artifacts repository

Zero-Downtime Rollouts: Nginx rolling reloads and systemd service updates on enterprise Linux VM

VERIFIED DEVOPS TECHNICAL COMPETENCY DOMAINS

CLOUD INFRASTRUCTURE
- Microsoft Azure
- VNet, Subnets, NSG, VM
- Storage, Key Vault
- Azure PostgreSQL Flex
- Landing Zone Concepts

INFRASTRUCTURE AS CODE
- Terraform Modules
- Remote State (Blob)
- Workspaces, Variables
- Checkov & tfsec Scans
- tflint & Formatting

CI/CD & VERSION CONTROL
- Azure DevOps Pipelines
- GitHub Actions
- YAML Multi-Stage
- Git & Pull Requests
- Branch Policies

LINUX & SCRIPTING
- RHEL & Ubuntu Admin
- systemd & Services
- Bash & Shell Scripting
- Python & FastAPI
- Nginx Reverse Proxy

MONITORING & SECURITY
- Prometheus Monitoring
- Azure Monitor & Alerts
- Secrets Management
- RBAC Least Privilege
- Shift-Left IaC Security

AUTOMATION & TOOLS
- Ansible Playbooks
- Power Automate Flows
- JIRA Delivery Tracking
- Confluence Docs
- Agile & DevOps Delivery

NETWORKING & TELECOM
- TCP/IP, DNS, Routing
- iptables & Firewalls
- VLANs, tcpdump, Wireshark
- 4G / 5G RAN Concepts
- NOC / BSS Operations